

Test 2: Monotonicity verification

Verdict: PASS | Generated: 2026-09-14 23:47:15 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

What this test verifies

Checks $b(N)$ does not increase along the refinement sequence ? an algebraic consistency property (binary verdict).

METHOD: Checks $b(N)$ does not increase along the refinement sequence ? an algebraic consistency property (binary verdict).

Result

Test 2 [HARDCORE pass 2]: 4 sub-checks, 0 failed, $b(50000)=1.2126$? PASS

‘plots/plot_01.png’ ? 1600x1000, ABPlotV23 engine (supersampled)

Plots

- ‘plots/animation.gif’ ? animated reveal of every plot of this test

Reproduce with: `julia ab_cloud_v23.jl --test 2 --no-two-pass`

Independent verification guide

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

(no log_comp entries)

FULL COMPUTATION LOG

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CONSOLE CAPTURE

TEST 2 (secondary, HARDCORE): Objection 1: $b(N)$ Convergence ? deep monotonicity audit

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S1 strict band: 0/39 pairs violate (limit 0)

segment	mean ??
1-5001	1.629872
5001-10001	1.328818
10001-15001	1.248992
15001-20001	1.201689
20001-25000	1.168653
25000-30000	1.143731
30000-35000	1.123510
35000-40000	1.106908
40000-45000	1.092713
45000-50000	1.080314

S2 segments : 0 local increases (limit 0)

S3 new lows : 40/40 checkpoints = 100% (limit ? 60%)

S4 drawdown : b(50000)=1.2126 vs b(25)=4.6101 ? 73.7% decrease (limit ? 5%)

??

S1 strict band PASS ? 0 violations / 39 pairs

S2 segments PASS ? 0 increases / 10 segments

S3 new lows PASS ? 100% of checkpoints

S4 drawdown PASS ? 73.7% decrease

Test 2 [HARDCORE pass 2]: 4 sub-checks, 0 failed, b(50000)=1.2126 ? PASS